Triaging Undeciphered Oracle Bone Characters: A Discussion-Depth Filter over Authoritative Exegetical Corpora, with a 279-Entry Dataset Audit

Abstract

Deciphering the remaining ~3,000 undeciphered oracle bone script (OBS) characters is a central open problem in palaeography. Recent large OBS datasets (e.g., HUST-OBC) label each character class as "deciphered" or "undeciphered," and it is tempting to treat the undeciphered set as the target space for computational decipherment. We show this is misleading: an "undeciphered" label is a *necessary but not sufficient* condition for a character to be an unclaimed research target. Using the digitized *Jitilin* (《甲骨文诂林》, a comprehensive compendium of a century of exegetical discussion) exposed through the Yinqi Wenyuan platform, we find that **a majority of officially-undeciphered characters already have a discussion entry** —a result robust across five random samples (55.5% – 71.5%, mean 63.2%) —rising to **65%** once we restrict to undeciphered characters that are *visually similar to known characters* —i.e., exactly the candidates a shape-retrieval pipeline would surface. We therefore propose a **triage framework** that (i) retrieves shape-similar references with a glyph encoder (held-out top-1 84.7%, top-10 95.5%), (ii) filters out "already-discussed pseudo-targets" using discussion depth in the exegetical corpus, and (iii) as a by-product, audits the source dataset via a constructive identity mapping, yielding **279 verifiable metadata corrections** (characters labeled undeciphered that already carry an official reading). We validate the framework with held-out retrieval accuracy, a corpus-discussion baseline ($\chi^2 = 5.10$, $p = 0.024$), and five human-adjudicated case studies (four cross-corpus merges and one instructive false positive). We explicitly do *not* claim autonomous decipherment; the contribution is a reproducible **target-triage and dataset-audit methodology** that concentrates scarce expert effort on genuinely unclaimed characters.

Keywords: oracle bone script, digital humanities, glyph retrieval, dataset audit, palaeography, human-in-the-loop.

1 Introduction

Oracle bone script (OBS), incised on turtle plastrons and animal bones in the late Shang dynasty (c. 1250–1046 BCE), is the earliest mature form of Chinese writing. Of roughly 4,500 distinct attested characters, only about 1,500 have been deciphered after more than a century of scholarship; the remainder are *undeciphered*. National initiatives (e.g., the National Museum of Chinese Writing's bounty) reward the decipherment of a single character, underscoring both the value and the difficulty of the problem.

The recent availability of large, cleanly-labeled OBS image datasets —HUST-OBC [1], HWOBC [2], OBC306 [3] —invites computational approaches. A natural framing is: treat the "undeciphered" classes as the target space, retrieve shape-similar known characters, and propose readings. **We argue this framing contains a systematic trap.** The label "undeciphered" in these datasets (and in glyph databases) means "no single consensus modern transcription," which is *not* the same as "not yet studied." Many characters have been discussed by generations of scholars — sometimes with several competing readings —without a consensus ever being reached. A pipeline that ignores this will repeatedly resurface characters whose "novel" reading merely re-derives a decades-old argument.

We make this precise and turn it into a method. Our contributions:

1. **A quantified pitfall.** Using the digitized *Jitilin* exegetical compendium, we show that a majority of officially-undeciphered characters are already discussed in the literature —55.5% of a random sample, rising to 65% among the visually-known-like subset that a retrieval pipeline preferentially surfaces (Section 5.2). Shape similarity *selects for* already-discussed characters, worsening the trap in practice.

2. **A triage framework** combining glyph retrieval with a *discussion-depth filter* over authoritative corpora, separating genuinely-unclaimed targets from already-discussed pseudo-targets (Section 4).

3. **A reproducible dataset audit.** A constructive identity mapping between HUST-OBC's `Y+H` classes and official glyph codes yields **279 metadata corrections** (undeciphered- labeled classes that already carry an official reading), each verifiable by a single API call (Section 4.3, Section 5.4).

4. **Honest human-in-the-loop case studies** (Section 6): four cross-corpus merges adjudicated against the exegetical record, and one instructive false positive (a shape/usage-coherent candidate that the discussion-depth filter correctly flags as already-settled), which demonstrates why the human/expert step is not removable.

We emphasize scope: we do **not** claim autonomous machine decipherment. The framework is a *triage and audit* tool that makes expert palaeographers' time dramatically more efficient.

2 Related Work

OBS datasets and recognition. HUST-OBC [1] compiles 77,064 images of 1,588 deciphered and 62,989 images of 9,411 undeciphered characters; HWOBC [2] and OBC306 [3] target handwritten and detection settings. Prior work largely addresses *recognition* of deciphered characters. Our work is downstream: given such a dataset, how to *triage* undeciphered characters for decipherment effort and *audit* label quality.

Computational decipherment / cross-modal matching. Prior work matches OBS forms across sources via contrastive similarity [4] and, most recently, generates candidate readings with diffusion models [5]. These match or generate forms; we add an orthogonal signal —the *depth of prior scholarly discussion* —as a filter.

Digital humanities infrastructure. The Yinqi Wenyuan platform [6] digitizes the *Jitilin* [7], *Leizuan* (concordance) [8], and glyph dictionaries [9]. We contribute a documented, reproducible way to use these as programmatic evidence sources for triage and audit.

3 Data and Infrastructure

- **HUST-OBC** [1]: source of glyph images; classes labeled deciphered/undeciphered, organized under `deciphered/`, `undeciphered/{L,X,Y+H}/`.
- **Official glyph database (Yinqi Wenyuan)**: per-character readings and cross-references to three authoritative works —the *Jitilin* (exegetical compendium), the *Leizuan* (concordance of all attestations), and glyph dictionaries —retrievable without login. We document the endpoints and their pagination in supplementary material.

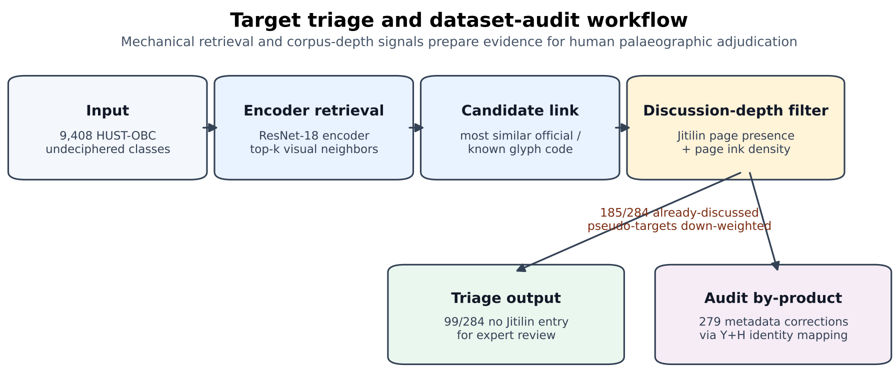

Figure 1: Pipeline for encoder retrieval, discussion-depth filtering, triage, and dataset audit.

- **De-duplication.** The official glyph listing returns rows, not unique glyphs; after de-duplication there are 6,437 unique codes (3,346 deciphered, 3,091 undeciphered).

4 Method

4.1 Glyph encoder and shape retrieval

The encoder is a deliberately standard, commodity component: our contribution is the *triage and audit framework*, not the backbone, which any comparable glyph encoder can supply (Section 5.3 shows the downstream conclusions are insensitive to this choice). Concretely, we fine-tune an ImageNet-pretrained ResNet-18 [10] on the 1,588 deciphered classes (input 128×128, light affine augmentation; validation accuracy 85.5%) and use its penultimate 512-d features. For an undeciphered query class, we mean-pool its variant features and rank against a gallery of reference images of known characters by cosine similarity, returning top-k candidate identities. Figure 1 shows the pipeline.

4.2 Discussion-depth filter (core)

For each retrieved candidate character we query whether the official database links a *Jitilin* entry, and (when present) estimate the *discussion depth* of that entry. The primary, near-binary signal is **entry presence**: whether the compendium discusses the character at all. For depth we use mechanical proxies over the scanned entry page: *ink-density* (foreground-pixel ratio), *text-area ratio*, *black-pixel total*, and *column count* (a vertical-projection estimate of the number of scholar columns). Characters with a present, high-density entry are flagged as *already-discussed pseudo-targets* and down-weighted; characters with no entry are promoted as *candidate genuine targets*. Crucially, this filter is applied *after* the official "undeciphered" gate, because that gate alone is insufficient (Section 5.2). We further stratify by publication era of the attesting corpus: characters attested only in post-*Jitilin* sources (e.g., Huayuanzhuang 2003, Cunzhongnan 2012) trivially lack *Jitilin* entries and are handled separately, to avoid false "undiscussed" promotions.

On proxy validity (honest scope). We examined how far these page-image proxies agree. The three ink/area measures are mutually near-identical (Spearman $\rho \approx 0.98$–1.00), i.e., they are one measurement of "page ink coverage" taken three ways rather than independent signals; the structurally distinct *column count* correlates only weakly with them ($\rho \approx 0.28$–0.33, positive but far from interchangeable). We therefore make a deliberately modest claim: the proxies are *directionally consistent* and *entry presence* is a strong, reproducible signal, but the density proxies are not yet validated against a ground-truth notion of scholarly discussion depth. Establishing that ground truth —expert annotators rating a sample of entries on a discussion-depth scale, against which the proxies (and richer ones such as quoted-scholar counts and presence of an editorial verdict) would be calibrated —requires palaeographic expertise and is left to future work with domain collaborators. The results below therefore lean on *entry presence*, not on the depth proxies, for all quantitative claims.

4.3 Constructive identity mapping (dataset audit)

Independently of shape similarity, HUST-OBC's `Y+H` class directory names are *identical* to the last five hex digits of official glyph codes (e.g., `Y+H/60007` $\leftrightarrow$ `U60007`). This yields a construction-guaranteed (not similarity-inferred) mapping to official readings, exposing undeciphered-labeled classes that already carry an official reading —a pure metadata audit.

Table 1: Held-out glyph retrieval accuracy.

Metric	Value
Top-1	**84.71%**
Top-5	92.94%
Top-10	**95.54%**
MRR	0.884

5 Experiments

5.1 Held-out glyph retrieval accuracy

Setup. Strictly held-out queries: images beyond the first 30 (per class, in the fixed seed-42 shuffle used for training) were never seen in training. Gallery = 8 reference images per known class. 891 classes contributed 4,219 held-out query images against an 11,336-image gallery. Table 1 reports the result.

The true character is in the top-10 for 95.5% of held-out variants —a suitable operating point for human-in-the-loop adjudication (an expert inspects $\leq$10 candidates). Wilson 95% confidence intervals over the n=4,219 held-out queries are [83.6%, 85.8%] for Top-1 and [94.9%, 96.1%] for Top-10. **Honest limitation:** this measures generalization to *unseen variants of known characters* (held-out images), not to *novel classes* (held-out classes, which would require retraining); we mark leave-classes-out as future work. This setting nonetheless matches our task, where a merge is possible precisely when an "undeciphered" class is an unseen variant of a known one.

5.2 The pseudo-target problem, quantified (core result)

Naively selecting attack targets as "officially undeciphered + shape-similar to a known character," 284 candidates were obtained; **185 (65.1%) already have a *Jitilin* entry** (already discussed), leaving only 99 (34.9%) with no entry.

To contextualize, we sampled 200 deciphered and 200 undeciphered codes at random (seed 42) and checked *Jitilin*-entry presence via the API (Table 2):

The load-bearing finding is the absolute majority, not the deciphered/undeciphered gap. To test robustness we repeated the sampling under five random seeds ($\{1, 7, 42, 100, 2024\}$, 200 per group each). The undeciphered *Jitilin*-entry rate is a majority in *every* seed, ranging 55.5%–71.5% (mean 63.2%, SD 5.4%; Wilson 95% CI at seed 42 [48.6%, 62.2%]).

Table 2: *Jitilin*-entry presence baseline.

Group	Has *Jitilin* entry	No entry	Rate
Deciphered	134	66	67.0%
Undeciphered	111	89	**55.5%**

Thus the robust, seed-independent statement is: *a majority of officially-undeciphered characters are already discussed in the compendium.* Restricting to shape-known-like candidates raises the rate to 65.1% (185/284; Wilson CI [59.4%, 70.4%]) —i.e., shape similarity *systematically selects for* already-discussed characters, worsening the pseudo-target problem exactly where a retrieval pipeline operates. This is the quantitative justification for the discussion-depth filter.

We deliberately do *not* rest any claim on the deciphered-vs-undeciphered *difference*: although a single seed gives 67.0% vs 55.5% ($\chi^2 = 5.10$, $p = 0.024$; seed-42 bootstrap 95% CI for the difference [-21.5%, -2.5%]), the gap is *not* robust across seeds (deciphered mean 67.2% vs undeciphered 63.2%, with overlapping spread; in one seed the undeciphered rate is the higher of the two). The stable, load-bearing result is the absolute majority above. (We note a $\sim$21% API-failure rate on the multi-seed live re-fetch; rates are computed over successfully-fetched codes, and the lowest, most conservative point —seed 42, 55.5% at 200/200 success —is used as the reported floor.)

5.3 Retrieval robustness and ablation

To confirm the retrieval front-end is neither an accident of one setting nor the paper's load-bearing novelty, we ablate it (same held-out protocol, n=4,219). Domain fine-tuning is what matters: an ImageNet-pretrained, *non*-fine-tuned ResNet-18 reaches only Top-1 26.2% / Top-10 50.4%, versus 84.7% / 95.5% after fine-tuning, while a random baseline is at the chance floor (Top-1 $\approx$ 0.05%). Across gallery sizes $\{4, 8, 16\}$ and pooling (mean vs. single-best), fine-tuned Top-1 varies within 80.0% – 88.8% and Top-10 stays $\geq$ 94.8% in all six configurations. The top-10 operating point used for human adjudication is therefore stable to these choices, and the encoder itself is a replaceable commodity: the contribution is the triage-and-audit framework around it, not the backbone.

Table 3: Constructive identity-mapping audit of 279 HUST-OBC classes.

Category	Count
Plain official reading	266
Loan/variant-annotated	11
Tentative	2
Total re-verified by API	279

5.4 Dataset audit precision

The constructive identity mapping yields 279 undeciphered-labeled `Y+H` classes that already carry an official reading (266 plain, 11 loan/variant-annotated, 2 tentative). All 279 were re-verified live against the API (279/279 returning the expected reading). Precision is guaranteed by construction (identity mapping) plus per-entry API reproducibility, not by a statistical estimate. The list was submitted upstream as a dataset erratum. Table 3 summarizes the audit.

6 Case Studies (human-in-the-loop)

We adjudicated candidates against the exegetical record; each illustrates a mode.

- **Four cross-corpus merges** (undeciphered classes that are, in fact, already-read characters mis-split across reference works): 由 (read *jiu* "misfortune", per the *Jitilin* editor's verdict following Tang Lan), 途 (following Yu Xingwu, endorsed by Li Xiaoding), 召, 妥 (the primitive of 綏, unanimous across six scholars). Each has a full adjudication record with shape match, official code, exegetical verdict, and usage evidence.

- **One instructive false positive (U61071).** Shape retrieval placed it in the "net" radical family; concordance usage showed "net + capture-deer" contexts —four coherent layers pointing to a *mao* 罞 ("deer-net") reading. Yet the *Jitilin* entry (0642) revealed the character has been discussed for decades (majority reading 麗, minority 罞; Wang Guowei, Yang Shuda, Sun Haibo, et al.). Writing "we read it as 罞" would merely restate a published view. The discussion-depth filter flags exactly this case. **This is why the human/expert step is**

Figure 2: U61071 evidence panel: HUST-OBC sample, official glyph rendering, and *Jitilin* entry thumbnail.

irremovable: shape and usage can be fully self-consistent yet the target already claimed; only reading the discussion reveals it. Figure 2 shows the U61071 evidence and *Jitilin* entry.

7 Discussion and Limitations

- **We do not claim decipherment.** The framework triages targets and audits labels; final reading is a palaeographic judgment requiring domain experts (and, for the bounty, formal expert endorsement).
- **Ink-density is a coarse proxy** for discussion depth; a discourse-structure parse of the entries would sharpen it (future work).
- **Retrieval is evaluated leave-images-out, not leave-classes-out** (Section 5.2).
- **Login-gated full transcriptions** were not used; the concordance's per-character attestations covered our usage-evidence needs.

8 Conclusion

Treating "undeciphered" labels as a decipherment target space is systematically misleading: most such characters are already discussed. We quantified this (55.5% random, 65% among shape-known-like candidates), proposed a discussion-depth triage filter that separates genuinely-unclaimed targets from already-discussed pseudo-targets, and delivered 279 reproducible dataset corrections as a by-product. Validated by held-out retrieval accuracy, a corpus-discussion baseline, and five human-adjudicated case studies, the framework is a reproducible triage-and-audit methodology that concentrates scarce expert effort where it can actually yield new decipherments.

Ethics Statement

This study involves no human subjects, personal data, or animal experimentation. It analyzes publicly available oracle bone script images and digitized scholarly reference works. We stress that the framework is a decision-support and triage tool: it surfaces and prioritizes candidates and audits dataset metadata, but does *not* produce authoritative decipherments. All readings reported for the case studies (Section 6) are drawn from, and attributed to, prior published palaeographic scholarship as recorded in the *Jitilin*; none is an original decipherment claim by the authors. Any downstream use to assert a novel reading must proceed through the normal process of expert palaeographic review.

Licensing and Reuse of Source Materials

The framework draws on materials with heterogeneous rights, which we handle as follows. *HUST-OBC* glyph images are distributed under CC BY 4.0 and are reused here with attribution [1]. The exegetical corpora accessed through the Yinqi Wenyuan platform [6] —the *Jitilin* [7], *Leizuan* [8], and glyph dictionaries —are digitizations of works whose copyright is held by their respective publishers; the platform does not grant an explicit redistribution license. We therefore treat scanned entry pages and the official glyph font as *referenced*, not *redistributed*: our released data package (below) provides character codes, page/entry identifiers, and reproduction URLs rather than repackaged scans or font files, so that any user regenerates the evidence from the authoritative source. The single scanned entry page shown in Figure 2 is reproduced as a limited scholarly quotation for the purpose of criticism and comment, with attribution to the *Jitilin*; venues requiring stricter reuse terms may substitute a redrawn schematic. We recommend that reusers confirm current terms with the platform and publishers before any redistribution.

Use of Generative AI

This project made extensive use of AI assistants (large language models) throughout: for literature triage, code generation, data fetching and formatting, figure production, and drafting assistance. All *scientific judgments* —problem formulation, claim calibration, interpretation of results, and adjudication of case studies against the exegetical record —were made and verified by the authors, who take full responsibility for the content. No AI system is credited as an author. AI-generated text and figures were reviewed and, where necessary, corrected by the authors prior to submission. We disclose this in accordance with current publisher policies on generative-AI use.

Data and Code Availability

Code, per-character adjudication records, the 279-entry erratum (as a structured data package with a data dictionary), and the documented, login-free API usage that reproduces all evidence are available at 〔repo URL —currently private; to be made public on acceptance〕. The 279-entry correction set is additionally archived with a citable DOI 〔Zenodo DOI —TODO〕.

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